THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · HARD LEVEL CHAPTER TEST SET 3

Chapter 1: **Chemical Reactions and Equations** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. In $Zn + CuSO_4 \longrightarrow ZnSO_4 + Cu$, the species reduced is: 1

(a) Zn **(b)** SO_4^{2-} **(c)** Cu^{2+} **(d)** Zn^{2+}

2. Assertion (A): The reaction between lead nitrate and potassium iodide is a double displacement reaction. Reason (R): It is a redox reaction. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

3. 5 g of A and 10 g of B react to give 12 g of C only. The mass of reactants left unreacted is: 1

(a) 15 g **(b)** 12 g **(c)** 5 g **(d)** 3 g

4. The best method to protect the iron hull of a ship is to attach blocks of: 1

(a) gold **(b)** magnesium or zinc
(c) silver **(d)** copper

5. Assertion (A): Electrolysis of water gives hydrogen and oxygen in the volume ratio 2 : 1. Reason (R): Water is a compound of hydrogen and oxygen. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

6. An open flask with reaction mixture loses mass from 50.0 g to 48.9 g because a gas escapes. The mass of gas lost is: 1

(a) 48.9 g **(b)** 0.9 g **(c)** 2.1 g **(d)** 1.1 g

7. A student writes $H_2 + O_2 \longrightarrow H_2O_2$ as the balanced form for water formation. The error is: 1

(a) coefficient is correct
(b) formula of hydrogen is wrong
(c) oxygen should not be 2
(d) changed the formula of the product, which is not allowed

8. Assertion (A): Burning of magnesium ribbon is a reduction reaction of magnesium. Reason (R): Magnesium gains oxygen during burning. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

9. In $N_2 + 3\,H_2 \longrightarrow 2\,NH_3$, 30 mL of H2 with excess N2 gives ammonia of volume: 1

(a) 30 mL **(b)** 60 mL **(c)** 20 mL **(d)** 10 mL

10. A white solid on heating gives a gas that turns lime water milky and a residue that gives slaked lime with water. The solid is: 1

(a) $NaCl$ **(b)** $CaCO_3$ **(c)** $CaCl_2$ **(d)** $CaSO_4$

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. A reaction mixture becomes warm as X reacts with Y. What type of reaction is it? Give one example. 2

12. Magnesium continues to burn in carbon dioxide. Write the equation and say which substance is oxidised. 2

13. A copper sulphate solution cannot be stored in an iron vessel. Explain with an equation. 2

14. In electrolysis of water 10 mL of oxygen is collected. Find the volume of hydrogen and justify it. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. 10 g of CaCO3 is heated (Ca = 40, C = 12, O = 16). Find the masses of CaO and CO2 formed. 3

16. Zinc displaces copper from copper sulphate but copper does not react with zinc sulphate. Explain using reactivity. 3

17. Compare the energy changes in the reactions of quick lime with water and decomposition of limestone. 3

OR

14.35 g of silver chloride decomposes in sunlight (Ag = 108, Cl = 35.5). Calculate the mass of silver obtained. 3

18. A student balances $H_2 + O_2 \longrightarrow H_2O$ by writing $H_2 + O_2 \longrightarrow H_2O_2$. Explain the error and correct it. 3

OR

Give the roles of 3
(i) oxygen in rusting,
(ii) moisture in rusting,
(iii) zinc in galvanisation.

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. For each reaction state oxidised, reduced and the reducing agent: 5
(i) $2\,Mg + O_2 \longrightarrow 2\,MgO$
(ii) $H_2S + Cl_2 \longrightarrow 2\,HCl + S$
(iii) $ZnO + C \longrightarrow Zn + CO$.

OR

Explain how galvanisation, painting, oiling and alloying prevent corrosion, and why sacrificial zinc plates are fixed on ship hulls. 5

20. Balance and classify: 5
(a) $Fe + O_2 \longrightarrow Fe_2O_3$
(b) $NH_3 + CuO \longrightarrow Cu + N_2 + H_2O$
(c) $Al + Fe_3O_4 \longrightarrow Al_2O_3 + Fe$.

OR

Write one equation each for thermal, electrolytic and photolytic decomposition, state the energy used in each and one application. 5

END OF QUESTION PAPER — Total: 40 Marks